Background

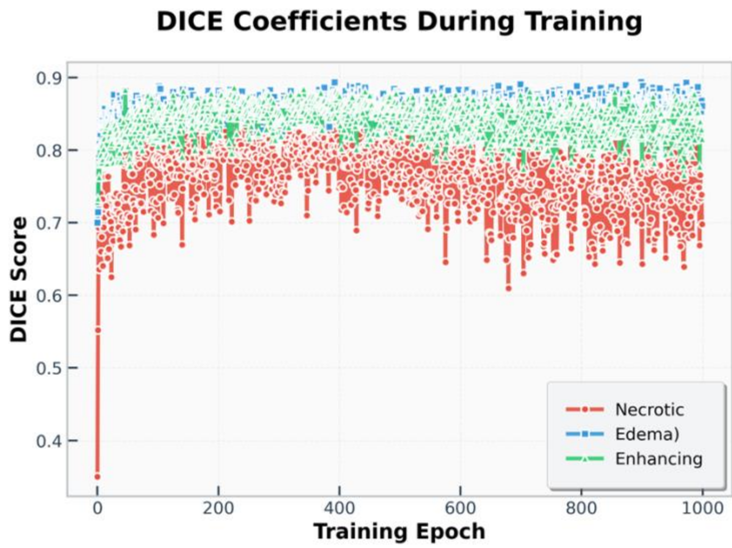
Glioblastoma (GBM) is a highly aggressive brain tumor with frequent recurrence and shows significant heterogeneity. Identifying multifocal recurrence remains a major clinical challenge.

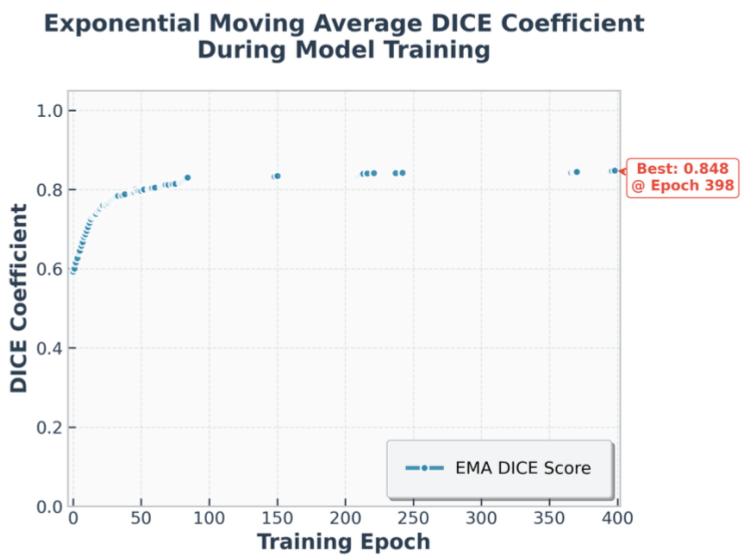
Methods

We developed a deep learning model trained on MRI scans of GBM patients and fine-tuned on multifocal GBM cases to enhance recognition of complex spatial and textural features. A secondary classifier was created to predict recurrence, distinguishing between multifocal and primary disease. Visualization of activation maps from encoder layers was used to interpret the learned features and assess their biological relevance.

[Add Network Architecture]

Training and validation losses





Metrics

class_id	Dice_mean	IoU_mean	Dice_median	IoU_median	Dice_std	IoU_std	FN_sum
1	0.68	0.56	0.75	0.60	0.23	0.24	164108
2	0.82	0.71	0.84	0.72	0.10	0.14	194291
3	0.83	0.72	0.84	0.73	0.10	0.13	47770

More Metrics (Jaccard Index, Hausdorff95)

Classifier metrics: primary vs. secondary tumors

[spatial heatmaps on your segmentation outputs]

Clinical and/or molecular data integration

Recurrence location prediction?

Results

Preliminary findings show that the model captures characteristic texture patterns associated with multifocal morphology and recurrence risk.

Conclusion

This framework demonstrates the potential of deep learning to identify multifocal recurrence in GBM and to provide interpretable imaging biomarkers that could guide patient monitoring and personalized treatment strategies.